

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

750 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

750 KWP GRID-TIED SOLAR PLANT

Net investment ₹2.71 Cr after input tax credit, generating 11,49,750 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.5 yrs

Post-tax, undiscounted

PROJECT IRR

29.0%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹2.98

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹5.48 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	750 kWp
Year 1 generation	11,49,750 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹82.32 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹2.85 Cr
GST at 5% — recovered as ITC	₹13.57 L
Net capitalised cost	₹2.71 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹68.31 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,71,42,857	1,08,57,143	1,62,85,714	27,32,526
2	1,62,85,715	65,14,286	97,71,429	16,39,516
3	97,71,428	39,08,571	58,62,857	9,83,709
4	58,62,857	23,45,143	35,17,714	5,90,226
5	35,17,715	14,07,086	21,10,629	3,54,135
6	21,10,628	8,44,251	12,66,377	2,12,481
7	12,66,377	5,06,551	7,59,826	1,27,489
8	7,59,827	3,03,931	4,55,896	76,493
9	4,55,895	1,82,358	2,73,537	45,896
10	2,73,537	1,09,415	1,64,122	27,538
11	1,64,122	65,649	98,473	16,523
12	98,473	39,389	59,084	9,913
Tax shield, years 1-12				68,16,444

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30-year post-tax cash flow

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How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	11,49,750	82,32,210	—	1,08,57,143	(6,60,643)	88,92,853	88,92,853
2	11,44,001	85,18,691	4,50,000	65,14,286	3,91,213	76,77,478	1,65,70,331
3	11,38,281	88,15,141	4,50,000	39,08,571	11,21,630	72,43,512	2,38,13,843
4	11,32,590	91,21,908	4,50,000	23,45,143	15,92,320	70,79,588	3,08,93,431
5	11,26,927	94,39,351	4,50,000	14,07,086	19,08,304	70,81,046	3,79,74,477
6	11,21,292	97,67,840	4,50,000	8,44,251	21,32,633	71,85,207	4,51,59,684
7	11,15,686	1,01,07,761	4,50,000	5,06,551	23,03,177	73,54,584	5,25,14,268
8	11,10,107	1,04,59,511	4,50,000	3,03,931	24,42,700	75,66,811	6,00,81,079
9	11,04,557	1,08,23,502	4,50,000	1,82,358	25,64,907	78,08,595	6,78,89,674
10	10,99,034	1,12,00,160	4,50,000	1,09,415	26,78,063	80,72,097	7,59,61,771
11	10,93,539	1,15,89,925	4,50,000	65,649	27,87,174	83,52,752	8,43,14,523
12	10,88,071	1,19,93,255	4,50,000	39,389	28,95,293	86,47,962	9,29,62,485
13	10,82,631	1,24,10,620	4,50,000	23,634	30,04,301	89,56,319	10,19,18,804
14	10,77,218	1,28,42,510	4,50,000	14,180	31,15,378	92,77,132	11,11,95,936
15	10,71,832	1,32,89,429	4,50,000	8,508	32,29,286	96,10,143	12,08,06,079
16	10,66,472	1,37,51,901	4,50,000	5,105	33,46,538	99,55,363	13,07,61,442
17	10,61,140	1,42,30,467	4,50,000	3,063	34,67,497	1,03,12,970	14,10,74,412
18	10,55,834	1,47,25,688	4,50,000	1,838	35,92,443	1,06,83,245	15,17,57,657
19	10,50,555	1,52,38,141	4,50,000	1,103	37,21,602	1,10,66,540	16,28,24,197
20	10,45,302	1,57,68,429	4,50,000	662	38,55,176	1,14,63,253	17,42,87,450
21	10,40,076	1,63,17,170	4,50,000	397	39,93,349	1,18,73,821	18,61,61,271
22	10,34,876	1,68,85,008	4,50,000	238	41,36,303	1,22,98,705	19,84,59,976
23	10,29,701	1,74,72,606	4,50,000	143	42,84,213	1,27,38,392	21,11,98,368
24	10,24,553	1,80,80,653	4,50,000	86	44,37,261	1,31,93,392	22,43,91,760
25	10,19,430	1,87,09,859	4,50,000	51	45,95,628	1,36,64,231	23,80,55,991
26	10,14,333	1,93,60,962	4,50,000	31	47,59,503	1,41,51,459	25,22,07,450
27	10,09,261	2,00,34,724	4,50,000	19	49,29,079	1,46,55,645	26,68,63,095
28	10,04,215	2,07,31,932	4,50,000	11	51,04,554	1,51,77,378	28,20,40,473
29	9,99,194	2,14,53,404	4,50,000	7	52,86,135	1,57,17,269	29,77,57,742
30	9,94,198	2,21,99,982	4,50,000	4	54,74,034	1,62,75,948	31,40,33,690
Total	3,21,04,656	42,35,72,740	1,30,50,000	2,71,42,853	9,64,89,051	31,40,33,690	31,40,33,690

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

750 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹5.48 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹31.40 Cr
30-year generation	3,21,04,655 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,21,04,655 units, avoiding 26,325.8 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹2.85 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 750 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.